

Qualification Pack



3D Animator (TV and Film Making)

QP Code: NFD/MES/Q3501

Version: 1.0

NSQF Level: 4

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NFD/MES/Q3501: 3D Animator (TV and Film Making)

Brief Job Description

This course is a short-term skill training program for freshers and amateurs to learn the craft of 3D animation filmmaking and earn a livelihood through employment or freelance assignments. The objective of the program is to prepare individuals for a job as 3D Animation Designer where they create visual concepts, using computer software or by hand, to communicate ideas that inspire, inform, and captivate consumers and enable them to develop the overall process and production design for various applications such as Film, Game & advertisements, TV series with high visual impact.

Personal Attributes

This job requires individuals with a creative flair and technical proficiency in 3D animation, capable of using computer software to craft compelling visual concepts. They should exhibit a keen eye for detail, an innovative approach to storytelling, and the ability to conceptualize and execute designs that resonate with audiences. These professionals must be adept at developing the overall process and production design for diverse applications like films, games, advertisements, and TV series, delivering high-impact visual experiences that engage and captivate viewers.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [NFD/MES/N3505: Carry out pre-production activities for 3D animation for TV and Film making.](#)
2. [NFD/MES/N3501: Carry out modeling, texturing and rigging for 3D animation.](#)
3. [NFD/MES/N3504: Carry out 3D animation.](#)
4. [NFD/MES/N3503: Carryout post-production for 3D animation](#)
5. [NFD/MES/N3502: Specialization in Pre-Production/3D Asset Creation/Animation/Post Production](#)
6. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Media & Entertainment
Sub-Sector	Animation and Gaming
Occupation	Animation

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Country	India
NSQF Level	4
Credits	18
Aligned to NCO/ISCO/ISIC Code	1330.0101, 2432.0201
Minimum Educational Qualification & Experience	12th grade Pass (with basic computer skills) with NA of experience OR Completed 2nd year of the 3-year diploma after 10 (in animation, computer graphics) with NA of experience
Minimum Level of Education for Training in School	12th Class
Pre-Requisite License or Training	No specialization required
Minimum Job Entry Age	17 Years
Last Reviewed On	NA
Next Review Date	31/10/2026
NSQC Approval Date	01/11/2023
Version	1.0
Reference code on NQR	2020/ME/NFDC03965
NQR Version	1

Remarks:

Not Applicable

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NFD/MES/N3505: Carry out pre-production activities for 3D animation for TV and Film making.

Description

This NOS is designed to perform pre-production activities for 3D animation for TV and Film making.

Scope

The scope covers the following :

- Pre-production activities for 3D animation for TV and Film making.

Elements and Performance Criteria

Carry out pre-production activities for 3D animation for TV and Film making

To be competent, the user/individual on the job must be able to:

- PC1.** develop your own story from given reference points.
- PC2.** write own script from story
- PC3.** identify the correct hardware and its use across stages of production
- PC4.** demonstrate original cartoon character from their imagination
- PC5.** demonstrate original storyboard of their own story
- PC6.** demonstrate process of cinematography
- PC7.** use different camera angles appropriately
- PC8.** apply the various types of visualization , storytelling techniques for Animation film making.
- PC9.** prepare own storyboard & Animatics from the original story

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the process of developing a compelling narrative for animation.
- KU2.** Ability to generate original stories based on provided reference points.
- KU3.** Knowledge of key elements in storytelling, such as character development, plot structure, and theme integration.
- KU4.** Proficiency in writing scripts based on the developed story.
- KU5.** Understanding of screenplay format and industry standards.
- KU6.** Ability to translate narrative ideas into a well-structured script suitable for animation production.
- KU7.** Identify and comprehend the correct hardware used in different stages of animation production.
- KU8.** Understand the functionality and application of hardware such as computers, graphic tablets, and specialized animation tools.
- KU9.** Demonstrate the ability to create original cartoon characters from imagination.

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- KU10.** Understanding of character anatomy, expressions, and movement in the context of animation.
- KU11.** Ability to create an original storyboard that visually represents the script.
- KU12.** Knowledge of industry-standard storyboard formats and terminology.
- KU13.** Proficiency in storytelling through sequential visual elements.
- KU14.** Understand the process of cinematography in animation production.
- KU15.** Knowledge of framing, composition, and camera movement in a virtual 3D space.
- KU16.** Ability to effectively use lighting, color, and camera angles to enhance the visual storytelling.
- KU17.** Demonstrate the use of different camera angles appropriately.
- KU18.** Understand the impact of camera angles on storytelling and audience engagement.
- KU19.** Proficiency in selecting and employing camera angles for specific scenes and moods.
- KU20.** Apply various types of visualization techniques relevant to animation film making.
- KU21.** Understand and utilize storytelling techniques unique to the animation medium.
- KU22.** Incorporate visual storytelling elements to convey emotions, atmosphere, and narrative effectively.
- KU23.** Ability to prepare a storyboard and animatics from the original story.
- KU24.** Proficiency in translating the script into a visual sequence of shots.
- KU25.** Understand the importance of animatics in refining pacing, timing, and overall narrative flow.
- KU26.** Stay updated on industry standards, regulations, and best practices related to animation in TV and film making.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Comprehend complex technical manuals, scripts, storyboards, and other production materials with accuracy and speed.
- GS2.** Create clear, concise, and grammatically correct written documents such as scripts, storyboards, animatics, and production notes.
- GS3.** Actively listen to and understand instructions, feedback, and ideas from clients, directors, and other team members.
- GS4.** Communicate effectively both verbally and non-verbally to present ideas, explain concepts, and collaborate with others.
- GS5.** Analyze information, identify and evaluate options, and make sound decisions considering technical limitations, artistic vision, and budgetary constraints.
- GS6.** Develop and manage production schedules, allocate resources effectively, and prioritize tasks to meet deadlines.
- GS7.** Understand and anticipate client needs, respond to feedback with flexibility, and ensure satisfaction with the final product.
- GS8.** Identify and troubleshoot technical, creative, and logistical issues that arise during the pre-production stage.
- GS9.** Break down complex problems into smaller manageable parts, analyze data, and draw logical conclusions to inform decision-making.

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- GS10.** Evaluate ideas objectively, identify potential flaws or inconsistencies in concepts, and propose constructive solutions.
- GS11.** Develop original story ideas, character designs, and visual concepts.
- GS12.** Interpret and translate written descriptions and concepts into visually appealing storyboards and animatics.
- GS13.** Operate 3D modeling, animation, and rendering software proficiently.
- GS14.** Effectively communicate and collaborate with artists, directors, producers, and other team members to achieve shared goals.
- GS15.** Work efficiently and manage time effectively to meet deadlines and maintain a productive workflow.
- GS16.** Pay close attention to detail throughout the pre-production process to ensure accuracy and quality in the final animation.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Carry out pre-production activities for 3D animation for TV and Film making</i>	40	40	20	-
PC1. develop your own story from given reference points.	-	-	-	-
PC2. write own script from story	-	-	-	-
PC3. identify the correct hardware and its use across stages of production	-	-	-	-
PC4. demonstrate original cartoon character from their imagination	-	-	-	-
PC5. demonstrate original storyboard of their own story	-	-	-	-
PC6. demonstrate process of cinematography	-	-	-	-
PC7. use different camera angles appropriately	-	-	-	-
PC8. apply the various types of visualization , storytelling techniques for Animation film making.	-	-	-	-
PC9. prepare own storyboard & Animatics from the original story	-	-	-	-
NOS Total	40	40	20	-

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National Occupational Standards (NOS) Parameters

NOS Code	NFD/MES/N3505
NOS Name	Carry out pre-production activities for 3D animation for TV and Film making.
Sector	Media & Entertainment
Sub-Sector	Animation and Gaming
Occupation	Animation
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	01/11/2023
Next Review Date	31/10/2026
NSQC Clearance Date	01/11/2023

Qualification Pack

NFD/MES/N3501: Carry out modeling, texturing and rigging for 3D animation.

Description

This NOS is designed to perform modeling, texturing and rigging for 3D animation.

Scope

The scope covers the following :

- Foundation of props & background modeling
- Character modeling Art
- UV wrapping and texturing
- Foundation of 3D Rigging

Elements and Performance Criteria

Foundation of props & background modeling

To be competent, the user/individual on the job must be able to:

- PC1.** Use the interface of maya modeling.
- PC2.** Create 3D flower Vase model in Maya
- PC3.** demonstrate maya tools & settings for 3D modeling.
- PC4.** create props model in 3d software
- PC5.** Students needs to show proper topology and low poly modeling process
- PC6.** Create a 3D background like exterior /interior . The 3D model must be low poly with proper topology

Character modeling Art

To be competent, the user/individual on the job must be able to:

- PC7.** create detailing of 3D topology for character modeling
- PC8.** create complete low poly character 3D model

UV wrapping and texturing

To be competent, the user/individual on the job must be able to:

- PC9.** cloth 3D models
- PC10.** carry out UV wrapping
- PC11.** carry out texturing and apply colors as per color theory
- PC12.** to submit one Props model & character model with texture

Foundation of 3D Rigging

To be competent, the user/individual on the job must be able to:

- PC13.** Demonstrate basic tools and settings required in Maya for rigging
- PC14.** Design setup for given 3D model for rigging
- PC15.** Implement proper technique on the assigned model to create rigging
- PC16.** Complete rigging for mechanical 3D props

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- PC17.** Demonstrate bone setup in human 3D model
- PC18.** Design controller and assign them to the human 3D model
- PC19.** Complete a human rigging
- PC20.** Demonstrate basic tools and settings required in Maya for rigging

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understanding of 3D modelling concepts like polygon modelling, NURBS, subdivision surfaces, retopology, low-poly modelling, and topology optimization.
- KU2.** Knowledge of different modelling techniques like box modelling, polygon modelling, edge looping, extruding, chamfering, beveling, and sculpting.
- KU3.** Familiarity with 3D modelling software interfaces and functionalities, specifically Maya.
- KU4.** Understanding of anatomy and proportions for creating realistic human and animal characters.
- KU5.** Knowledge of character topology for optimal rigging and deformation.
- KU6.** Skills in facial sculpting and expressions for creating lifelike characters.
- KU7.** Understanding of UV unwrapping techniques for optimal texture mapping.
- KU8.** Knowledge of texture painting tools and techniques, including color theory and texture blending.
- KU9.** Familiarity with different texture maps like diffuse, normal, specular, roughness, and displacement maps.
- KU10.** Understanding of rigging principles and terminology, including bones, joints, IK/FK controllers, skin weights, and deformers.
- KU11.** Knowledge of different rigging techniques for various character types and props.
- KU12.** Skills in creating efficient and flexible rigs for animation purposes.
- KU13.** Expertise in navigating the Maya interface and utilizing its modelling, UV unwrapping, texturing, and rigging tools.
- KU14.** Ability to customize Maya preferences and hotkeys for efficient workflow.
- KU15.** Proficient in applying various modelling techniques to create props, backgrounds, and characters.
- KU16.** Understanding of polygon optimization and low-poly modelling for efficient rendering.
- KU17.** Skills in performing unwrapping techniques like automatic unwrapping, seam placement, and manual unwrapping for optimal texture coverage.
- KU18.** Ability to create high-quality textures using painting tools and procedural textures.
- KU19.** Proficiency in rigging different types of characters and props using bones, joints, controllers, and skin weights.
- KU20.** Ability to troubleshoot rigging issues and optimize rigs for animation performance.
- KU21.** Understanding of production pipelines for 3D animation projects.
- KU22.** Ability to work collaboratively with other artists and animators within a team.
- KU23.** Meeting deadlines and adhering to project specifications.
- KU24.** Maintaining effective communication with supervisors and clients.

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- KU25.** Staying updated on the latest trends and technologies in 3D animation.
- KU26.** Basic knowledge of other 3D animation software like Blender or ZBrush.
- KU27.** Understanding of animation principles and concepts for creating believable character movement.
- KU28.** Strong problem-solving skills and ability to adapt to changing project requirements.
- KU29.** Time management and organizational skills to meet deadlines effectively.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Write clear and concise reports, instructions, and emails.
- GS2.** Effectively communicate design ideas and technical concepts to colleagues and clients.
- GS3.** Present work in a clear and organized manner during meetings and presentations.
- GS4.** Actively listen to and understand feedback from colleagues and clients.
- GS5.** Identify and analyze problems and errors in 3D models.
- GS6.** Develop creative solutions to overcome technical challenges.
- GS7.** Think critically about design decisions and their impact on the final product.
- GS8.** Test and refine 3D models to ensure they meet quality standards.
- GS9.** Adapt to changing project requirements and deadlines.
- GS10.** Plan and prioritize tasks effectively to meet deadlines.
- GS11.** Manage time effectively to complete multiple projects simultaneously.
- GS12.** Organize and maintain digital assets (models, textures, etc.) for easy access.
- GS13.** Break down complex tasks into smaller, manageable steps.
- GS14.** Collaborate effectively with other team members (artists, animators, directors).
- GS15.** Make informed decisions based on technical knowledge, artistic principles, and project requirements.
- GS16.** Evaluate different options and choose the best solution for the project.
- GS17.** Weigh risks and benefits before making decisions.
- GS18.** Take responsibility for the quality of your work.
- GS19.** Be open to feedback and willing to adjust your approach when necessary.
- GS20.** Understand the needs and expectations of clients and stakeholders.
- GS21.** Communicate effectively with clients to ensure their satisfaction.
- GS22.** Meet deadlines and deliver high-quality work that meets client specifications.
- GS23.** Be open to feedback and willing to make changes to your work based on client input.
- GS24.** Build strong relationships with clients and colleagues.
- GS25.** Proficiency in industry-standard 3D modeling software (Maya, Blender, etc.).
- GS26.** Strong understanding of 3D modeling principles (topology, low poly modeling, etc.).
- GS27.** Skill in UV unwrapping and texturing.
- GS28.** Knowledge of fundamental 3D rigging techniques.
- GS29.** Ability to create realistic and aesthetically pleasing 3D models and textures.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Foundation of props & background modeling</i>	10	10	5	-
PC1. Use the interface of maya modeling.	-	-	-	-
PC2. Create 3D flower Vase model in Maya	-	-	-	-
PC3. demonstrate maya tools & settings for 3D modeling.	-	-	-	-
PC4. create props model in 3d software	-	-	-	-
PC5. Students needs to show proper topology and low poly modeling process	-	-	-	-
PC6. Create a 3D background like exterior /interior . The 3D model must be low poly with proper topology	-	-	-	-
<i>Character modeling Art</i>	10	10	5	-
PC7. create detailing of 3D topology for character modeling	-	-	-	-
PC8. create complete low poly character 3D model	-	-	-	-
<i>UV wrapping and texturing</i>	10	10	5	-
PC9. cloth 3D models	-	-	-	-
PC10. carry out UV wrapping	-	-	-	-
PC11. carry out texturing and apply colors as per color theory	-	-	-	-
PC12. to submit one Props model & character model with texture	-	-	-	-
<i>Foundation of 3D Rigging</i>	10	10	5	-
PC13. Demonstrate basic tools and settings required in Maya for rigging	-	-	-	-
PC14. Design setup for given 3D model for rigging	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. Implement proper technique on the assigned model to create rigging	-	-	-	-
PC16. Complete rigging for mechanical 3D props	-	-	-	-
PC17. Demonstrate bone setup in human 3D model	-	-	-	-
PC18. Design controller and assign them to the human 3D model	-	-	-	-
PC19. Complete a human rigging	-	-	-	-
PC20. Demonstrate basic tools and settings required in Maya for rigging	-	-	-	-
NOS Total	40	40	20	-

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National Occupational Standards (NOS) Parameters

NOS Code	NFD/MES/N3501
NOS Name	Carry out modeling, texturing and rigging for 3D animation.
Sector	Media & Entertainment
Sub-Sector	Animation and Gaming
Occupation	Animation
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	01/11/2023
Next Review Date	31/10/2026
NSQC Clearance Date	01/11/2023

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NFD/MES/N3504: Carry out 3D animation.

Description

This NOS is designed to perform 3D animation.

Scope

The scope covers the following :

- Fundamental of basic Animation
- Introduction to 3D Character animation
- Body mechanics and Character performance
- Acting for animation and animatics to animation

Elements and Performance Criteria

Fundamental of basic Animation

To be competent, the user/individual on the job must be able to:

- PC1.** Successfully showcase rolling ball animation for implementing animation principles.
- PC2.** Use proper process of keyframing and timeline navigation
- PC3.** Demonstrate basic of timing & spacing for animation principle
- PC4.** Show object weight by using animation principle
- PC5.** Demonstrate timing and spacing in key frame animation in maya
- PC6.** Showcase how to animate different types of objects to confirm their weight
- PC7.** Demonstrate Follow through & overlapping properly
- PC8.** Use proper method to showcase squash & stretch in 3D animation

Introduction to 3D Character animation

To be competent, the user/individual on the job must be able to:

- PC9.** Demonstrate of animation tools & setting in Maya
- PC10.** Put keyframe and use timeline in Maya
- PC11.** Navigate Maya interface by using available options.
- PC12.** Successfully showcase matching process of video footage and 3D rig character.
- PC13.** Use proper process to import motion capture data in Maya timeline and clean up keyframes
- PC14.** Taking output from Maya for rotomation animation
- PC15.** Demonstrate blocking of human walk cycle
- PC16.** Showcase process of keyframing and posing for walk & cycle
- PC17.** Show proper distribution of key frame in the timeline for human walk cycle
- PC18.** Demonstrate graph editor polishing for walk cycle
- PC19.** Demonstrate blocking of human run cycle
- PC20.** Carry our proper distribution of key frame in the timeline for human run cycle
- PC21.** Demonstrate graph editor polishing for run cycle

Body mechanics & character performance

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To be competent, the user/individual on the job must be able to:

PC22. Demonstrate different human action pose making ability from given reference

PC23. Design character key poses for jump animation

PC24. Perform Adding of breakdown

PC25. Complete Jump animation with proper graph & dope sheet

PC26. Design key poses of weight lift animation

PC27. Demonstrate proper blocking & breakdown process of weight lift animation

PC28. Finalize weight lift animation

PC29. Demonstrate complete process of push animation

Acting for animation and animatics to animation

To be competent, the user/individual on the job must be able to:

PC30. Demonstrate proper technique of character posing

PC31. Showcase acting ability and shoot by video camera

PC32. Implement audio in Maya

PC33. Character blocking in maya from video acting reference

PC34. Implement breakdown and in between

PC35. Showcase proper process of graph editor & dope sheet

PC36. Adding camera and showing play blast output

PC37. Deliver proper blocking from animatics

PC38. Design facial lip sync in maya with audio

PC39. Create lip sync animation in animatics to animation & acting from animation

PC40. Implement breakdown and in between

PC41. Showcase proper process of graph editor & dope sheet

PC42. Adding camera and showing play blast output

PC43. Demonstrate basic expression poses in maya from given reference

PC44. Show human eye blink in maya

PC45. Design different expressions in Maya

PC46. Demonstrate proper human facial animation with expressions in Maya

PC47. Prepare Basic object animation, walk run, human body mechanics & Character Acting

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand and apply animation principles in the creation of a rolling ball animation.

KU2. Demonstrate knowledge of keyframing and timeline navigation processes.

KU3. Exhibit a grasp of the basics of timing and spacing principles in animation.

KU4. Demonstrate the application of timing and spacing in keyframe animation using Maya.

KU5. Showcase the understanding of object weight through animation principles.

KU6. Demonstrate the ability to animate various objects to convey realistic weight.

KU7. Showcase Follow-through & Overlapping effectively in animations.

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- KU8.** Apply the principles of squash and stretch appropriately in 3D animation.
- KU9.** Exhibit proficiency in using animation tools and settings within the Maya software.
- KU10.** Showcase the skill of putting keyframes and utilizing the timeline in Maya.
- KU11.** Navigate the Maya interface proficiently using available options.
- KU12.** Successfully match video footage with 3D rigged characters.
- KU13.** Import motion capture data into the Maya timeline and perform keyframe cleanup.
- KU14.** Generate rotomation animation output from Maya.
- KU15.** Demonstrate the blocking process for a human walk cycle.
- KU16.** Showcase keyframing and posing techniques for a realistic walk cycle.
- KU17.** Properly distribute keyframes in the timeline for a human walk cycle.
- KU18.** Demonstrate graph editor polishing for a refined walk cycle.
- KU19.** Execute the blocking process for a human run cycle.
- KU20.** Distribute keyframes in the timeline for a human run cycle.
- KU21.** Demonstrate graph editor polishing for a dynamic run cycle.
- KU22.** Demonstrate the ability to create various human action poses from given references.
- KU23.** Design key poses for jump animation and execute the animation process.
- KU24.** Design key poses and perform blocking and breakdown processes for weight lift animation.
- KU25.** Finalize weight lift animation with proper techniques.
- KU26.** Implement proper techniques for character posing.
- KU27.** Showcase acting ability and capture animations using a video camera.
- KU28.** Integrate audio into Maya animations effectively.
- KU29.** Implement character blocking in Maya based on video acting references.
- KU30.** Incorporate breakdown and in-between frames effectively.
- KU31.** Showcase the proper utilization of the graph editor and dope sheet.
- KU32.** Add cameras and present playblast output.
- KU33.** Design facial lip sync in Maya using audio.
- KU34.** Create lip sync animations from animatics to final animation.
- KU35.** Implement breakdown and in-between frames for facial animations.
- KU36.** Showcase the process in the graph editor and dope sheet.
- KU37.** Add cameras and present playblast output.
- KU38.** Demonstrate the ability to create basic expression poses in Maya from references.
- KU39.** Animate human eye blinks effectively.
- KU40.** Design various expressions in Maya.
- KU41.** Showcase proper human facial animation with expressions.
- KU42.** Showcase proficiency in basic object animation, walk/run cycles, human body mechanics, and character acting.

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.** Demonstrate creative thinking and artistic flair in showcasing animations, understanding and implementing animation principles effectively.
- GS2.** Exhibit a meticulous approach in the keyframing and timeline navigation process, ensuring precision and accuracy in animations.
- GS3.** Communicate effectively through animations, conveying object weight, timing, and spacing principles with clarity.
- GS4.** Showcase the ability to interpret and translate given references into compelling animations.
- GS5.** Demonstrate proficiency in using animation tools and settings in Maya.
- GS6.** Apply technical knowledge in importing motion capture data, cleaning up keyframes, and utilizing rotomation animation techniques.
- GS7.** Adapt to different animation requirements by showcasing the ability to animate various objects, walk cycles, run cycles, and action poses with diverse styles.
- GS8.** Exhibit problem-solving skills in the animation process, especially in the blocking and breakdown stages for walk cycles, run cycles, and specialized animations.
- GS9.** Efficiently manage time in the creation of animations, including the design of key poses, breakdowns, and finalizing animations for jump, weight lift, and push actions.
- GS10.** Collaborate effectively in the animation process by showcasing the ability to match video footage with 3D rigged characters and integrating audio into Maya animations.
- GS11.** Demonstrate a willingness to learn and adapt to new techniques and tools in 3D animation, staying updated with industry trends and technology.
- GS12.** Pay close attention to the nuances of human movement, ensuring proper distribution of keyframes, and utilizing the graph editor for polishing walk and run cycles.
- GS13.** Showcase the ability to incorporate filmic elements, such as camera angles, playblast outputs, and proper blocking from animatics, to enhance the overall cinematic quality of animations.
- GS14.** Demonstrate the ability to convey emotions and expressions effectively through facial animations, including lip sync, eye blinks, and different expressions in Maya.
- GS15.** Display versatility in executing basic object animations, walk/run cycles, human body mechanics, and character acting, ensuring a well-rounded skill set in 3D animation.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Fundamental of basic Animation</i>	10	10	5	-
PC1. Successfully showcase rolling ball animation for implementing animation principles.	-	-	-	-
PC2. Use proper process of keyframing and timeline navigation	-	-	-	-
PC3. Demonstrate basic of timing & spacing for animation principle	-	-	-	-
PC4. Show object weight by using animation principle	-	-	-	-
PC5. Demonstrate timing and spacing in key frame animation in maya	-	-	-	-
PC6. Showcase how to animate different types of objects to confirm their weight	-	-	-	-
PC7. Demonstrate Follow through & overlapping properly	-	-	-	-
PC8. Use proper method to showcase squash & stretch in 3D animation	-	-	-	-
<i>Introduction to 3D Character animation</i>	10	10	5	-
PC9. Demonstrate of animation tools & setting in Maya	-	-	-	-
PC10. Put keyframe and use timeline in Maya	-	-	-	-
PC11. Navigate Maya interface by using available options.	-	-	-	-
PC12. Successfully showcase matching process of video footage and 3D rig character.	-	-	-	-
PC13. Use proper process to import motion capture data in Maya timeline and clean up keyframes	-	-	-	-
PC14. Taking output from Maya for rotomation animation	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. Demonstrate blocking of human walk cycle	-	-	-	-
PC16. Showcase process of keyframing and posing for walk & cycle	-	-	-	-
PC17. Show proper distribution of key frame in the timeline for human walk cycle	-	-	-	-
PC18. Demonstrate graph editor polishing for walk cycle	-	-	-	-
PC19. Demonstrate blocking of human run cycle	-	-	-	-
PC20. Carry out proper distribution of key frame in the timeline for human run cycle	-	-	-	-
PC21. Demonstrate graph editor polishing for run cycle	-	-	-	-
<i>Body mechanics & character performance</i>	10	10	5	-
PC22. Demonstrate different human action pose making ability from given reference	-	-	-	-
PC23. Design character key poses for jump animation	-	-	-	-
PC24. Perform Adding of breakdown	-	-	-	-
PC25. Complete Jump animation with proper graph & dope sheet	-	-	-	-
PC26. Design key poses of weight lift animation	-	-	-	-
PC27. Demonstrate proper blocking & breakdown process of weight lift animation	-	-	-	-
PC28. Finalize weight lift animation	-	-	-	-
PC29. Demonstrate complete process of push animation	-	-	-	-
<i>Acting for animation and animatics to animation</i>	10	10	5	-
PC30. Demonstrate proper technique of character posing	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC31. Showcase acting ability and shoot by video camera	-	-	-	-
PC32. Implement audio in Maya	-	-	-	-
PC33. Character blocking in maya from video acting reference	-	-	-	-
PC34. Implement breakdown and in between	-	-	-	-
PC35. Showcase proper process of graph editor & dope sheet	-	-	-	-
PC36. Adding camera and showing play blast output	-	-	-	-
PC37. Deliver proper blocking from animatics	-	-	-	-
PC38. Design facial lip sync in maya with audio	-	-	-	-
PC39. Create lip sync animation in animatics to animation & acting from animation	-	-	-	-
PC40. Implement breakdown and in between	-	-	-	-
PC41. Showcase proper process of graph editor & dope sheet	-	-	-	-
PC42. Adding camera and showing play blast output	-	-	-	-
PC43. Demonstrate basic expression poses in maya from given reference	-	-	-	-
PC44. Show human eye blink in maya	-	-	-	-
PC45. Design different expressions in Maya	-	-	-	-
PC46. Demonstrate proper human facial animation with expressions in Maya	-	-	-	-
PC47. Prepare Basic object animation, walk run, human body mechanics & Character Acting	-	-	-	-
NOS Total	40	40	20	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NFD/MES/N3504
NOS Name	Carry out 3D animation.
Sector	Media & Entertainment
Sub-Sector	Animation and Gaming
Occupation	Animation
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	01/11/2023
Next Review Date	31/10/2026
NSQC Clearance Date	01/11/2023

Qualification Pack

NFD/MES/N3503: Carryout post-production for 3D animation

Description

This NOS is designed to perform post-production for 3D animation

Scope

The scope covers the following :

- Shading network and materials
- 3D Lighting & CGI light magic
- Rendering and setting preference
- Process of basic compositing
- Fundamental of Editing & Final Showreel

Elements and Performance Criteria

Shading network and materials

To be competent, the user/individual on the job must be able to:

- PC1.** create shaders in 3D model
- PC2.** present look development process by applying different maps on 3D object
- PC3.** Apply maya hyper shade
- PC4.** demonstrate different shaders like Lambert, phong, Blinn etc.

3D Lighting & CGI light magic

To be competent, the user/individual on the job must be able to:

- PC5.** create lighting on 3D props, character and background model
- PC6.** Create ambient light for a 3D background model.
- PC7.** Create studio & 3 point lighting for 3D character model
- PC8.** create shadows of different 3D models through lighting.

Rendering and setting preference

To be competent, the user/individual on the job must be able to:

- PC9.** demonstrate different 3D image render from Maya
- PC10.** Demonstrate the process to the different types of render via a Batch Render or via a Render Sequence.

Process of basic compositing

To be competent, the user/individual on the job must be able to:

- PC11.** Demonstrate the process of compositing.
- PC12.** create video output from different image sequence
- PC13.** Demonstrate complete process of color correction in after-effects
- PC14.** Apply the composition software criteria

Fundamental of Editing & Final Showreel

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC15.** submit proper Animation showreel with only best approved work with sound.
- PC16.** prepare showreel that is within the specified duration and quality parameters specified by client
- PC17.** prepare short film based on Story approved by client

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understanding the principles of shader creation in 3D modeling.
- KU2.** Knowledge of shader types and their applications in TV and film production.
- KU3.** Proficiency in creating and modifying shaders to achieve desired visual effects.
- KU4.** Ability to present the look development process by applying different maps on 3D objects.
- KU5.** Knowledge of texture mapping, bump mapping, and displacement mapping for enhanced realism.
- KU6.** Understanding the role of look development in creating visually appealing 3D models.
- KU7.** Proficiency in using Maya Hypershade for shader network creation and management.
- KU8.** Understanding the interface and functionalities of Maya Hypershade.
- KU9.** Ability to organize and optimize shader networks for efficient rendering.
- KU10.** Knowledge and demonstration of various shaders such as Lambert, Phong, Blinn, etc.
- KU11.** Understanding the properties and applications of different shaders in 3D animation.
- KU12.** Proficiency in selecting and applying appropriate shaders based on project requirements.
- KU13.** Ability to create effective lighting setups for 3D props, characters, and background models.
- KU14.** Knowledge of the impact of lighting on mood, tone, and storytelling in TV and film.
- KU15.** Proficiency in balancing different light sources to achieve desired visual effects.
- KU16.** Understanding the concept of ambient light in 3D modeling for background scenes.
- KU17.** Ability to create ambient light setups that enhance the overall atmosphere of a 3D background model.
- KU18.** Proficiency in adjusting ambient light intensity and color temperature.
- KU19.** Knowledge and application of studio lighting principles for 3D character models.
- KU20.** Ability to set up three-point lighting to highlight key features and create depth.
- KU21.** Proficiency in adjusting light angles and intensities for optimal character presentation.
- KU22.** Ability to create realistic shadows for different 3D models using lighting.
- KU23.** Understanding the impact of light direction and source characteristics on shadow formation.
- KU24.** Proficiency in adjusting shadow parameters for desired visual effects.
- KU25.** Knowledge and demonstration of the process of 3D image rendering in Maya.
- KU26.** Understanding render settings, resolution, and format options for image output.
- KU27.** Proficiency in producing high-quality still images from 3D scenes.
- KU28.** Ability to demonstrate the process of different types of rendering via Batch Render or Render Sequence.
- KU29.** Knowledge of rendering settings and batch processing for efficient rendering workflows.
- KU30.** Proficiency in managing render queues and optimizing render times.

Qualification Pack

- KU31.** Knowledge and demonstration of the compositing process in post-production.
- KU32.** Understanding the integration of 3D renders with live-action footage or other elements.
- KU33.** Proficiency in using compositing software to achieve seamless visual integration.
- KU34.** Ability to create video output from different image sequences.
- KU35.** Knowledge of video format options, codecs, and compression techniques.
- KU36.** Proficiency in ensuring smooth playback and optimal video quality.
- KU37.** Demonstration of the complete process of color correction in Adobe After Effects.
- KU38.** Understanding color grading techniques to enhance the visual appeal of animations.
- KU39.** Proficiency in adjusting color balance, contrast, and saturation for desired effects.
- KU40.** Application of criteria for selecting and using composition software.
- KU41.** Knowledge of industry-standard composition software and their features.
- KU42.** Proficiency in choosing the appropriate composition software based on project requirements.
- KU43.** Ability to submit a proper animation showreel with only approved best works.
- KU44.** Understanding the importance of sound integration in an animation showreel.
- KU45.** Proficiency in adhering to specified duration and quality parameters set by the client.
- KU46.** Preparation of a short film based on a story approved by the client.
- KU47.** Knowledge of the entire film production process, including scripting, storyboarding, and editing.
- KU48.** Proficiency in managing project timelines and resources for successful short film production.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to think creatively and bring artistic expression to 3D models, shaders, and lighting setups.
- GS2.** Innovation in developing unique looks and styles for characters, props, and backgrounds.
- GS3.** Demonstrated capability to enhance visual appeal through creative choices in animation and lighting.
- GS4.** Meticulous attention to detail in the creation of shaders, textures, and lighting setups.
- GS5.** Precision in adjusting parameters to achieve realistic and visually appealing 3D models.
- GS6.** Consistent focus on refining and perfecting the details in the animation and rendering process.
- GS7.** Strong problem-solving skills in addressing challenges related to shaders, lighting, and rendering.
- GS8.** Proficiency in troubleshooting issues that may arise during the 3D animation and rendering process.
- GS9.** Ability to find creative solutions to technical problems and optimize workflows.
- GS10.** Effective communication of ideas, concepts, and progress to team members and stakeholders.
- GS11.** Clear articulation of the rationale behind chosen shaders, lighting setups, and animation decisions.

Qualification Pack

- GS12.** Collaborative communication to ensure alignment with client expectations and project requirements.
- GS13.** Efficient time management to meet project deadlines and adhere to production schedules.
- GS14.** Prioritization of tasks to optimize workflow and ensure timely completion of animation projects.
- GS15.** Ability to manage multiple aspects of the animation pipeline concurrently.
- GS16.** Adaptability to evolving project requirements and changes in client preferences.
- GS17.** Flexibility in incorporating feedback and making adjustments to meet quality standards.
- GS18.** Capacity to learn and apply new techniques, tools, and software as needed.
- GS19.** Strong technical skills in using 3D modeling software, specifically Maya.
- GS20.** Proficiency in applying various shaders (Lambert, Phong, Blinn, etc.) and creating realistic lighting.
- GS21.** Technical competence in rendering, compositing, and video output processes.
- GS22.** Collaboration with other team members, including modelers, animators, and compositors.
- GS23.** Teamwork in achieving a cohesive and unified visual style across different aspects of the animation.
- GS24.** Openness to feedback and constructive criticism for continuous improvement.
- GS25.** Effective project management skills in organizing and executing animation projects.
- GS26.** Ability to plan and execute the creation of a short film based on client-approved stories.
- GS27.** Oversight of timelines, resource allocation, and coordination with team members.
- GS28.** Client-focused approach to meet the specified duration and quality parameters.
- GS29.** Attentiveness to client-approved stories and the incorporation of client feedback.
- GS30.** Commitment to delivering animation showreels that align with client expectations.
- GS31.** Proficiency in integrating sound into animation showreels for a complete audio-visual experience.
- GS32.** Synchronization of sound elements with animation to enhance storytelling and engagement.
- GS33.** Understanding the role of sound in creating immersive and impactful animation.
- GS34.** Commitment to delivering high-quality animations and adhering to industry standards.
- GS35.** Rigorous quality assurance processes in the creation of shaders, lighting, and rendering.
- GS36.** Continuous evaluation and improvement of animation techniques for optimal visual outcomes

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Shading network and materials</i>	8	8	4	-
PC1. create shaders in 3D model	-	-	-	-
PC2. present look development process by applying different maps on 3D object	-	-	-	-
PC3. Apply maya hyper shade	-	-	-	-
PC4. demonstrate different shaders like Lambert, phong, Blinn etc.	-	-	-	-
<i>3D Lighting & CGI light magic</i>	8	8	4	-
PC5. create lighting on 3D props, character and background model	-	-	-	-
PC6. Create ambient light for a 3D background model.	-	-	-	-
PC7. Create studio & 3 point lighting for 3D character model	-	-	-	-
PC8. create shadows of different 3D models through lighting.	-	-	-	-
<i>Rendering and setting preference</i>	8	8	4	-
PC9. demonstrate different 3D image render from Maya	-	-	-	-
PC10. Demonstrate the process to the different types of render via a Batch Render or via a Render Sequence.	-	-	-	-
<i>Process of basic compositing</i>	8	8	4	-
PC11. Demonstrate the process of compositing.	-	-	-	-
PC12. create video output from different image sequence	-	-	-	-
PC13. Demonstrate complete process of color correction in after-effects	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Apply the composition software criteria	-	-	-	-
<i>Fundamental of Editing & Final Showreel</i>	8	8	4	-
PC15. submit proper Animation showreel with only best approved work with sound.	-	-	-	-
PC16. prepare showreel that is within the specified duration and quality parameters specified by client	-	-	-	-
PC17. prepare short film based on Story approved by client	-	-	-	-
NOS Total	40	40	20	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NFD/MES/N3503
NOS Name	Carryout post-production for 3D animation
Sector	Media & Entertainment
Sub-Sector	Animation and Gaming
Occupation	Animation
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	01/11/2023
Next Review Date	31/10/2026
NSQC Clearance Date	01/11/2023

Qualification Pack

NFD/MES/N3502: Specialization in Pre-Production/3D Asset Creation/Animation/Post Production

Description

This NOS is designed to perform specialization in Pre-Production/3D Asset Creation/Animation/Post Production.

Scope

The scope covers the following :

- Specialization in Pre-Production/3D Asset Creation/Animation/Post Production

Elements and Performance Criteria

Pre-Production: Crafting the visual blueprint via design elements and storyboarding, laying the groundwork for compelling visual narratives in media production.

To be competent, the user/individual on the job must be able to:

- PC1.** Develop original character concepts based on storyboards and the director's input.
- PC2.** Create character sheets with detailed anatomy, personality, and costume design specifications.
- PC3.** Adapt existing character designs for different media formats (e.g., film, TV, video games).
- PC4.** Design visually engaging and story-supporting backgrounds for various scenes.
- PC5.** Utilise perspective, colour theory, and lighting to create believable and immersive environments.
- PC6.** Adapt existing background designs for different environments and camera angles.
- PC7.** Design functional and visually appealing props that fit the narrative and style of the production.
- PC8.** Create functional and visually appealing props that effectively contribute to storytelling, demonstrating a comprehensive understanding of material selection, size constraints, production limitations, and overall narrative impact.
- PC9.** Creatively morph versatile props into scene-specific storytellers, breathing life into diverse characters through transformation
- PC10.** Bring stories to life! Masterfully weave scripts and narratives into captivating visual sequences through the magic of storyboarding.
- PC11.** Master the language of storytelling! Express camera angles, character movement, and key actions through crystal-clear visuals.
- PC12.** Bridge the gap between vision and execution! Collaborate with directors and writers to sculpt storyboards into masterful storytelling tools.
- PC13.** Bring static narratives to life! Breathe movement into storyboards by crafting simple animations that capture timing and flow.
- PC14.** Unleash the magic of animation! Utilize basic rigs and techniques to showcase character movement and interaction with dynamic flair.

Qualification Pack

PC15. Be the architect of improvement! Sharpen storyboards and narratives by providing insightful feedback based on animatic tests.

Asset Creation: Crafting 3D assets through character, background, prop modelling, texturing, and rigging to enrich visual storytelling in media production.

To be competent, the user/individual on the job must be able to:

PC16. Demonstrate proficiency in industry-standard 3D modeling software by constructing complex character, background, and prop models that accurately reflect project specifications.

PC17. Apply proper mesh topology and geometry principles to ensure models are optimized for animation and deformation, maintaining structural integrity and flexibility.

PC18. Achieve high fidelity in 3D models by meticulously adhering to the established style and design guidelines of the project, ensuring visual consistency and artistic coherence.

PC19. Employ various texturing techniques, including material assignment, shader manipulation, and procedural methods, to imbue 3D models with visually rich and impactful textures.

PC20. Develop advanced texturing skills to create realistic and aesthetically pleasing textures that enhance the overall detail, depth, and visual quality of 3D models.

PC21. Demonstrate an understanding of lighting principles and environmental factors when applying textures to 3D models, ensuring textures respond realistically to various lighting conditions and contribute to the narrative context. Create rigs for characters and props that enable smooth and efficient animation.

PC22. Implement appropriate joint hierarchies, constraints, and controls for various types of movement.

PC23. Test and refine rigs to ensure accuracy and flexibility for animators.

Mastering animation: Bringing life to characters, props, and scenes through expert movement and seamless sequencing for impactful storytelling.

To be competent, the user/individual on the job must be able to:

PC24. Animate characters to convey emotions, actions, and personality traits effectively.

PC25. Apply principles of animation like weight, balance, and anticipation to create believable movement.

PC26. Synchronise character animation with dialogue, audio effects, and environment dynamics.

PC27. Animate props realistically, considering their physical properties and interaction with characters.

PC28. Utilise dynamic simulations and physics constraints for enhanced believability of prop movement.

PC29. Integrate prop animation seamlessly into the overall scene animation.

PC30. Maintain consistency of character and prop animation across multiple scenes and sequences.

PC31. Apply principles of pacing and storytelling to create impactful and engaging animation sequences.

PC32. Collaborate with editors and directors to refine the timing and flow of animated scenes.

Mastering lighting, rendering, compositing, and special effects in post-production crafts impactful visual content across diverse media.

To be competent, the user/individual on the job must be able to:

PC33. Set up realistic and atmospheric lighting for various scenes and environments.

PC34. Different lighting techniques (e.g., key light, fill light, backlight).

PC35. Collaborate with directors and cinematographers to achieve

Qualification Pack

- PC36.** Choose appropriate rendering settings to achieve the desired visual quality and optimize render time.
- PC37.** Manage and prioritize rendering jobs to meet deadlines and project requirements.
- PC38.** Troubleshoot and resolve rendering errors and imperfections.
- PC39.** Combine rendered animation layers, visual effects, and background elements to create the final image.
- PC40.** Apply colour correction, grading, and other adjustments to achieve a consistent and visually appealing look.
- PC41.** Ensure seamless integration of animation, effects, and other elements for a believable and polished final product.
- PC42.** Integrate 2D and 3D special effects to enhance storytelling and visuals.
- PC43.** Utilize particle systems, simulations, and dynamics to create realistic and dynamic effects.
- PC44.** Implement special effects within budget and time constraints without compromising quality.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understanding the process of developing original character concepts based on storyboards and director input.
- KU2.** Knowledge of visual storytelling elements that contribute to character design, including anatomy, personality, and costume specifications.
- KU3.** Ability to adapt existing character designs for different media formats, considering the specific requirements of film, TV, and video games.
- KU4.** Knowledge of designing visually engaging and story-supporting backgrounds for various scenes.
- KU5.** Understanding the application of perspective, color theory, and lighting in creating believable and immersive environments.
- KU6.** Adaptation of background designs for different environments and camera angles.
- KU7.** Ability to design functional and visually appealing props that fit the narrative and style of the production.
- KU8.** Understanding of material selection, size constraints, production limitations, and overall narrative impact in prop creation.
- KU9.** Creativity in morphing versatile props into scene-specific storytellers, contributing to character development and storytelling.
- KU10.** Mastery of weaving scripts and narratives into captivating visual sequences through storyboarding.
- KU11.** Expression of camera angles, character movement, and key actions through clear visuals.
- KU12.** Collaboration with directors and writers to sculpt storyboards into masterful storytelling tools, bringing static narratives to life KU13 through simple animations.
- KU13.** Proficiency in industry-standard 3D modeling software for constructing character, background, and prop models.
- KU14.** Application of proper mesh topology and geometry principles for optimized animation and deformation.

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- KU15.** Adherence to established style and design guidelines, ensuring visual consistency and artistic coherence.
- KU16.** Development of advanced texturing skills, including material assignment, shader manipulation, and procedural methods.
- KU17.** Creation of realistic and aesthetically pleasing textures that enhance the overall detail, depth, and visual quality of 3D models.
- KU18.** Understanding of lighting principles and environmental factors when applying textures to ensure realism in various lighting conditions.
- KU19.** Creation of rigs for characters and props enabling smooth and efficient animation.
- KU20.** Implementation of appropriate joint hierarchies, constraints, and controls for various types of movement.
- KU21.** Testing and refining rigs to ensure accuracy and flexibility for animators.
- KU22.** Animation of characters to convey emotions, actions, and personality traits effectively.
- KU23.** Application of principles of animation like weight, balance, and anticipation for believable movement.
- KU24.** Synchronization of character animation with dialogue, audio effects, and environment dynamics.
- KU25.** Set up realistic and atmospheric lighting for various scenes and environments.
- KU26.** Knowledge of different lighting techniques (e.g., key light, fill light, backlight).
- KU27.** Collaboration with directors and cinematographers to achieve the desired visual atmosphere and storytelling impact.
- KU28.** Choice of appropriate rendering settings to achieve desired visual quality and optimize render time.
- KU29.** Management and prioritization of rendering jobs to meet deadlines and project requirements.
- KU30.** Troubleshooting and resolution of rendering errors and imperfections.
- KU31.** Combination of rendered animation layers, visual effects, and background elements for the final image.
- KU32.** Application of color correction, grading, and adjustments for a consistent and visually appealing look.
- KU33.** Seamless integration of animation, effects, and other elements to create a polished final product.
- KU34.** Integration of 2D and 3D special effects to enhance storytelling and visuals.
- KU35.** Utilization of particle systems, simulations, and dynamics for realistic and dynamic effects.
- KU36.** Implementation of special effects within budget and time constraints without compromising quality.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to develop original character concepts and adapt existing designs for various media formats.

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- GS2.** Innovative approach to designing visually engaging backgrounds and props that contribute to storytelling.
- GS3.** Creativity in morphing versatile props into scene-specific storytellers and bringing static narratives to life through animation.
- GS4.** Mastery in storyboarding, expressing camera angles, character movement, and key actions clearly.
- GS5.** Collaboration with directors and writers to sculpt storyboards into masterful storytelling tools.
- GS6.** Skill in breathing movement into storyboards through crafting simple animations capturing timing and flow.
- GS7.** Adaptation of character and background designs for different environments, media formats, and camera angles.
- GS8.** Flexibility in incorporating feedback and refining designs to meet project specifications.
- GS9.** Ability to integrate 2D and 3D special effects to enhance storytelling and visuals.
- GS10.** Effective collaboration with directors, writers, editors, and cinematographers to refine animation sequences.
- GS11.** Clear communication of design ideas, concepts, and feedback to team members.
- GS12.** Collaboration with editors and directors to refine the timing and flow of animated scenes.
- GS13.** Analytical thinking in the development of character sheets, considering anatomy, personality, and costume design specifications.
- GS14.** Problem-solving in the creation of functional and visually appealing props within material constraints.
- GS15.** Critical thinking in providing insightful feedback for animatic tests to improve storyboards and narratives.
- GS16.** Meticulous attention to detail in the construction of 3D models, including proper mesh topology and geometry principles.
- GS17.** Detail-oriented approach to achieving high fidelity in 3D models, adhering to established style and design guidelines.
- GS18.** Precise application of various texturing techniques to imbue 3D models with visually rich and impactful textures.
- GS19.** Efficient time management in the creation of character concepts, storyboards, and 3D models.
- GS20.** Prioritization of tasks to meet project deadlines and adhere to production schedules.
- GS21.** Management and prioritization of rendering jobs to meet deadlines and project requirements.
- GS22.** Proficiency in industry-standard 3D modeling software for character, background, and prop models.
- GS23.** Technical competence in rigging characters and props for smooth and efficient animation.
- GS24.** Understanding of lighting principles and different techniques for realistic and atmospheric lighting.
- GS25.** Rigorous testing and refinement of rigs to ensure accuracy and flexibility for animators.
- GS26.** Consistent maintenance of consistency in character and prop animation across multiple scenes and sequences.

Qualification Pack

- GS27.** Continuous quality assurance in rendering, troubleshooting errors, and achieving a polished final product.
- GS28.** Adherence to established style and design guidelines for achieving visual consistency and artistic coherence.
- GS29.** Consideration of size constraints, production limitations, and overall narrative impact in the creation of props.
- GS30.** Implementation of special effects within budget and time constraints without compromising quality.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Pre-Production: Crafting the visual blueprint via design elements and storyboarding, laying the groundwork for compelling visual narratives in media production.</i>	10	10	5	-
PC1. Develop original character concepts based on storyboards and the director's input.	-	-	-	-
PC2. Create character sheets with detailed anatomy, personality, and costume design specifications.	-	-	-	-
PC3. Adapt existing character designs for different media formats (e.g., film, TV, video games).	-	-	-	-
PC4. Design visually engaging and story-supporting backgrounds for various scenes.	-	-	-	-
PC5. Utilise perspective, colour theory, and lighting to create believable and immersive environments.	-	-	-	-
PC6. Adapt existing background designs for different environments and camera angles.	-	-	-	-
PC7. Design functional and visually appealing props that fit the narrative and style of the production.	-	-	-	-
PC8. Create functional and visually appealing props that effectively contribute to storytelling, demonstrating a comprehensive understanding of material selection, size constraints, production limitations, and overall narrative impact.	-	-	-	-
PC9. Creatively morph versatile props into scene-specific storytellers, breathing life into diverse characters through transformation	-	-	-	-
PC10. Bring stories to life! Masterfully weave scripts and narratives into captivating visual sequences through the magic of storyboarding.	-	-	-	-
PC11. Master the language of storytelling! Express camera angles, character movement, and key actions through crystal-clear visuals.	-	-	-	-
PC12. Bridge the gap between vision and execution! Collaborate with directors and writers to sculpt storyboards into masterful storytelling tools.	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Bring static narratives to life! Breathe movement into storyboards by crafting simple animations that capture timing and flow.	-	-	-	-
PC14. Unleash the magic of animation! Utilize basic rigs and techniques to showcase character movement and interaction with dynamic flair.	-	-	-	-
PC15. Be the architect of improvement! Sharpen storyboards and narratives by providing insightful feedback based on animatic tests.	-	-	-	-
<i>Asset Creation: Crafting 3D assets through character, background, prop modelling, texturing, and rigging to enrich visual storytelling in media production.</i>	10	10	5	-
PC16. Demonstrate proficiency in industry-standard 3D modeling software by constructing complex character, background, and prop models that accurately reflect project specifications.	-	-	-	-
PC17. Apply proper mesh topology and geometry principles to ensure models are optimized for animation and deformation, maintaining structural integrity and flexibility.	-	-	-	-
PC18. Achieve high fidelity in 3D models by meticulously adhering to the established style and design guidelines of the project, ensuring visual consistency and artistic coherence.	-	-	-	-
PC19. Employ various texturing techniques, including material assignment, shader manipulation, and procedural methods, to imbue 3D models with visually rich and impactful textures.	-	-	-	-
PC20. Develop advanced texturing skills to create realistic and aesthetically pleasing textures that enhance the overall detail, depth, and visual quality of 3D models.	-	-	-	-
PC21. Demonstrate an understanding of lighting principles and environmental factors when applying textures to 3D models, ensuring textures respond realistically to various lighting conditions and contribute to the narrative context. Create rigs for characters and props that enable smooth and efficient animation.	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. Implement appropriate joint hierarchies, constraints, and controls for various types of movement.	-	-	-	-
PC23. Test and refine rigs to ensure accuracy and flexibility for animators.	-	-	-	-
<i>Mastering animation: Bringing life to characters, props, and scenes through expert movement and seamless sequencing for impactful storytelling.</i>	10	10	5	-
PC24. Animate characters to convey emotions, actions, and personality traits effectively.	-	-	-	-
PC25. Apply principles of animation like weight, balance, and anticipation to create believable movement.	-	-	-	-
PC26. Synchronise character animation with dialogue, audio effects, and environment dynamics.	-	-	-	-
PC27. Animate props realistically, considering their physical properties and interaction with characters.	-	-	-	-
PC28. Utilise dynamic simulations and physics constraints for enhanced believability of prop movement.	-	-	-	-
PC29. Integrate prop animation seamlessly into the overall scene animation.	-	-	-	-
PC30. Maintain consistency of character and prop animation across multiple scenes and sequences.	-	-	-	-
PC31. Apply principles of pacing and storytelling to create impactful and engaging animation sequences.	-	-	-	-
PC32. Collaborate with editors and directors to refine the timing and flow of animated scenes.	-	-	-	-
<i>Mastering lighting, rendering, compositing, and special effects in post-production crafts impactful visual content across diverse media.</i>	10	10	5	-
PC33. Set up realistic and atmospheric lighting for various scenes and environments.	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC34. Different lighting techniques (e.g., key light, fill light, backlight).	-	-	-	-
PC35. Collaborate with directors and cinematographers to achieve	-	-	-	-
PC36. Choose appropriate rendering settings to achieve the desired visual quality and optimize render time.	-	-	-	-
PC37. Manage and prioritize rendering jobs to meet deadlines and project requirements.	-	-	-	-
PC38. Troubleshoot and resolve rendering errors and imperfections.	-	-	-	-
PC39. Combine rendered animation layers, visual effects, and background elements to create the final image.	-	-	-	-
PC40. Apply colour correction, grading, and other adjustments to achieve a consistent and visually appealing look.	-	-	-	-
PC41. Ensure seamless integration of animation, effects, and other elements for a believable and polished final product.	-	-	-	-
PC42. Integrate 2D and 3D special effects to enhance storytelling and visuals.	-	-	-	-
PC43. Utilize particle systems, simulations, and dynamics to create realistic and dynamic effects.	-	-	-	-
PC44. Implement special effects within budget and time constraints without compromising quality.	-	-	-	-
NOS Total	40	40	20	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NFD/MES/N3502
NOS Name	Specialization in Pre-Production/3D Asset Creation/Animation/Post Production
Sector	Media & Entertainment
Sub-Sector	Animation and Gaming
Occupation	Animation
NSQF Level	4
Credits	7
Version	1.0
Last Reviewed Date	01/11/2023
Next Review Date	31/10/2026
NSQC Clearance Date	01/11/2023

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/training center based on this criterion.
6. To pass the Qualification Pack, every trainee should score a minimum of 70% of aggregate marks to successfully clear the assessment.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NFD/MES/N3505.Carry out pre-production activities for 3D animation for TV and Film making.	40	40	20	-	100	15
NFD/MES/N3501.Carry out modeling, texturing and rigging for 3D animation.	40	40	20	-	100	20
NFD/MES/N3504.Carry out 3D animation.	40	40	20	-	100	25
NFD/MES/N3503.Carryout post-production for 3D animation	40	40	20	-	100	25
NFD/MES/N3502.Specialization in Pre-Production/3D Asset Creation/Animation/Post Production	40	40	20	-	100	10
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	5
Total	220	230	100	-	550	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.